MOHAMMAD ABU-SHELBAIA

mabushelbaia@gmail.com
(+970) 059-415-6035

mabushelbaia.com
github.com/mabushelbaia
Ramallah, Palestine

SUMMARY

Computer Engineering graduate with excellence from Birzeit University, now a Teaching and Research Assistant there. Focused on embedded systems and low-level software: ARM Cortex-M4, STM32 and assembly. Experienced in Python backend development and automation.

SKILLS

Languages: Python, LaTeX, Markdown, Bash, Assembly, Java, C, SQL, HTML, CSS

Technologies: Git, FastAPI, Flask, Asyncio, Django, Tailwind CSS, Docker, PostgreSQL, Arduino, Nginx, Gunicorn

EXPERIENCE

9/2024 – Present
Teaching And Research Assistant — Birzeit University
Assisted in delivering laboratory sessions on microcontroller programming and embedded systems. Provided mentorship and support to students, evaluated assignments, and contributed to the preparation of lab materials.

7/2024 – 12/2024
Backend Developer — Silal
Contributed to the development and maintenance of a data processing pipeline for managing and analyzing product data. Wrote unit tests to ensure the reliability of Flask API endpoints.

3/2024 – 7/2024
Backend Developer Intern — Silal
Built a full-stack bookstore application using Django and Tailwind CSS, implementing core functionality and an intuitive user interface. Deployed the application with Docker Compose, using Nginx, Gunicorn, and PostgreSQL for a robust and production-ready setup.

PROJECTS

Web, VitePress
PWA, CI/CD
ECE Labs Platform — birzeit.me
Web platform that replaces the static PDF lab manuals of the Birzeit ECE department: searchable pages, copy-paste code, and fixes that go live on merge through GitHub Actions. Installable for offline use, with printable PDFs built from LaTeX.

Embedded Systems
PIC16F877A, Assembly
PIC16F877A Calculator — github.com
Wrote a calculator in PIC assembly for the PIC16F877A, with single-button input and a 16x2 LCD display, designed and simulated in MPLAB and Proteus.

Security, Cryptography
CBC, Docker
Padding Oracle Attack — github.com
Implemented a practical padding oracle attack against CBC mode with PKCS#5 padding, recovering the plaintext byte by byte, and packaged it to run in Docker.

Web Scraping
Python, iCal
Course Schedule — github.com
Built a Python CLI that logs into Ritaj, fetches the course schedule, and exports recurring iCal events merged with academic calendar dates such as holidays and exam periods.

EDUCATION

9/2020 - 7/2024
Birzeit University — Bachelor
Computer Engineering
`Cumulative Average: 87%`

28th July 2024
Udacity — Nanodegree Certificate
Data Analyst
`Certificate Link`

LANGUAGES

English - C2 (CEFR), **Arabic** - Native